More efficient plants, less electricity?

Assignment 2: Evaluating an energy-consulting programme

Economic Analysis for Data Scientists – Pau Milán (BSE)

Groups of three; approximately eight hours of group work. Submit one self-contained R Jupyter notebook. Question 9 and the electricity-levy extension are optional and outside the core workload.

Your assignment

A government offers energy consulting to manufacturing plants, hoping to reduce electricity consumption. You are the data-science team evaluating the pilot. What does the evidence show? Can the theory from Unit 3 help explain firms' behaviour and inform the government's next decision?

The data follow **480 plants for 16 monthly observations**, from month -3 through month 12. Consulting begins in month 1. Half the plants in each sector and staffing-review cohort are randomly assigned to consulting; the rest receive no consulting during the panel. Output quality and each plant's output price, wage and electricity price remain constant. Quantities contain small recording errors. Assume no spillovers between plants.

Research context. This assignment is inspired by **Nicholas Ryan (2018), "Energy Productivity and Energy Demand: Experimental Evidence from Indian Manufacturing Plants," NBER Working Paper 24619**, and the May 7, 2018 author manuscript. Working-paper record | Author manuscript. Our data and institutional details are synthetic, and our two-input model is a teaching simplification. The time path illustrates a scenario motivated by the paper's model, not a replication of its experimental time-series evidence. Reading the paper is not required to begin.

The data

In `data/energy_firms.csv`, one row is one plant-month. Output is in standardized batches of unchanged quality; labor is in **blocks of 100 paid worker-hours**; electricity is in **MWh**. All quantities are monthly totals; prices are in fictional currency units (CU).

Variable(s)	Meaning
`firm_id`, `sector`	Plant identifier; Textiles or Chemicals.
`treated`, `month`	Consulting assignment (1 or 0); month relative to programme start.
`output`, `labor`, `electricity`	Observed quantities q, L and E.
`output_price`, `wage`, `energy_price`	p, w, c: CU per batch, labor block and MWh.
`adjust_month`, `labor_adjustable`	Staffing-review month and adjustment indicator; see the institutional context in Question 4.

1 Explore electricity use before and after consulting

Explore and check the data. Present evidence about these plants' electricity consumption before and after the programme begins. You might examine averages, distributions, individual trajectories or changes relative to each plant's baseline. There is no single correct presentation: choose the comparisons and visualisations you find informative, and explain what they reveal.

As part of your exploration, check the sample, panel structure, units and treatment assignment. Note any data issues or important differences across plants. You do not need to propose an economic explanation yet.

Find the question, then develop an explanation

2 What does the time pattern tell you?

Look more closely at the evolution of electricity use. Does a single average for the period after consulting give an adequate account? Identify the features of the time path that would matter to a policymaker, and decide how to present them clearly.

Compare your conclusions across treated and control plants and consider whether the average hides differences across firms. State the main empirical pattern that your subsequent analysis needs to explain. Keep a description of that pattern separate from an explanation of why it occurs.

3 Evaluate the programme

Use the random assignment to assess whether consulting achieved its stated objective. Explain your comparison, your choice of outcome and time windows, and why the comparison is informative about the programme rather than simply about differences across plants or changes over time.

Report the size and uncertainty of the effects you consider relevant. Treat plants, rather than their repeated monthly observations, as the independent assignment units. Does the policy assessment depend on the horizon? Be precise about the difference between an average plant's proportional response and the change in total electricity consumption.

4 Use Unit 3 to make sense of the findings

Imagine that plant i has the technology

$$q = A_i \sqrt{\min\{L, \eta E\}}.$$

$A_i > 0$ is persistent plant productivity; $\eta > 0$ measures the energy services obtained from physical electricity, in units comparable with labor. Consulting aims to improve η. Neither its initial level nor its realised percentage change is supplied. Plants take their observed prices as given and seek to maximise operating profits, $pq - wL - cE$.

Institutional context. Before the intervention plants could choose all inputs. From month 1, pre-existing staffing commitments fix paid labor until the recorded adjustment month (4, 6 or 8). From that month onward labor can adjust freely; electricity can always adjust. These dates were assigned independently of plant characteristics before treatment assignment, and controls face the same arrangements. Assume no other variable inputs, inventory accumulation, entry or exit.

Use the theory learned in Unit 3 to construct an explanation for the findings so far and how they vary across horizons. State the assumptions your argument requires and support it with the calculations and economic diagrams you judge necessary. Decide which choices and constraints matter at each horizon; do not stop at a verbal story loosely connected to the model.

Explain what your account would lead you to look for elsewhere in the data. Distinguish properties assumed by the technology from conclusions supported by observed behaviour. You do not need to estimate a substitution elasticity.

Before the TA session
Make an initial attempt at Questions 1–3. Bring the pattern you want to explain and your proposed evaluation. The TA will help you develop the economic reasoning; the handout deliberately does not provide a sequence of algebraic steps.

Turn your explanation into further evidence

5 Can the initial response reveal the efficiency gain?

The programme assigns consulting, not a known percentage increase in efficiency. Using your explanation from Question 4, can you use the observations before the intervention and in **months 1–3** to estimate the technological effect of consulting?

Explain which observed quantities you need and why. In particular, establish the conditions under which the initial electricity response is informative about efficiency, and check whether the accompanying observations support those conditions. Distinguish the level of η from its percentage change, and use the control plants to assess the change attributable to consulting. Allow for recording noise by combining observations where appropriate.

Then investigate whether the efficiency change you infer appears to persist later. Explain how you would assess this once other input quantities can change. Be clear about which parts of the conclusion depend on the proposed technology.

6 Does your explanation predict more than the initial pattern?

A convincing mechanism should generate additional, testable implications. Derive predictions from your explanation and choose new tables, figures or comparisons to investigate each of the following.

(a) **The magnitude of later choices.** Does profit maximisation predict plants' later output and input quantities accurately? Use only baseline observations and months 1–3 to recover any required plant parameters. Make predictions before comparing them with the held-out observations in months 9–12. Examine discrepancies as well as fit.

(b) **Production and electricity.** Is the observed change in production large enough to account for the later electricity response, given the efficiency change you estimated? Derive a quantitative criterion and identify which plants meet it. A matching sign alone is not sufficient.

(c) **Differences across firms.** According to your mechanism, which pre-intervention characteristics should predict the strongest later electricity responses? Derive and investigate that prediction. Distinguish differences in the size of the efficiency improvement from differences in firms' responses to a given improvement.

(d) **Input relationships and timing.** What relationships between labor and electricity changes should your explanation imply? Do these differ when you compare later observations with baseline rather than with the initial post-intervention period? Investigate these relationships and whether the timing of responses is consistent with the institutional setting.

Explain what each piece of evidence establishes. Distinguish predictions tested against additional observations from identities built from the same quantities. Consider at least one alternative explanation, what would distinguish it from yours, and any limits of the available evidence.

Use the model to advise

7 Evaluate a further efficiency improvement

Suppose a follow-up programme could raise each treated plant's current η by a further **20%**, with A_i and prices unchanged. This is additional to the original intervention; there are no observations from the proposed programme.

Predict output, labor, electricity and operating profits with labor first fixed at the current medium-run level, then freely adjustable. Compare both scenarios with the same no-follow-up baseline. Aggregate the results and examine differences across plants. Explain why extrapolating the original percentage effect might be misleading. Profits exclude the unknown cost of delivering consulting.

Optional policy extension. After the original intervention, what plant-specific electricity levy would keep medium-run electricity use at or below its pre-intervention level? Set the levy to zero where none is needed. Let plants reoptimise; report output, profits and tax revenue. Explain why this is not an estimate of an optimal environmental tax or necessarily a feasible uniform policy.

8 Brief the policymaker

Write a 200–300 word briefing assessing the original programme and the proposed follow-up. Explain the mechanism in plain language, ground your recommendation in evidence and uncertainty, and identify the most useful additional information or policy test.

Distinguish electricity savings from output, private profits and social welfare. Environmental damages and programme-delivery costs are not observed. There is no prescribed policy recommendation.

9 Optional: a short Cobb–Douglas comparison

Consider the alternative $q = A_i L^{1/4} (\eta E)^{1/4}$. At unchanged input prices, derive its optimal labor-to-electricity ratio when both inputs can adjust. With labor fixed, determine the direction of the electricity response to improved efficiency. How do these predictions compare with the minimum model and the evidence?

This is a short comparison to clarify how technology affects behaviour. **Do not recalibrate or simulate the alternative.** Explain why rejecting it would not establish that the minimum technology is uniquely correct. This question is outside the core workload.

What to hand in

Submit a self-contained R Jupyter notebook with selected outputs, economic diagrams, calculations and written interpretation. Keep the supplied CSV in a `data` folder beside it; use relative paths and check that all cells run. State units, sample sizes, time windows and the comparison behind each effect. Plant-level uncertainty intervals are sufficient; advanced panel econometrics is not required.

Select the evidence that matters and explain how it supports your argument. Identify group members' contributions and any optional work. Assessment covers the evaluation, economic reasoning, mechanism evidence, counterfactuals and reproducibility. Full marks are available from Questions 1–8; optional work is not required.